
ABHISHEK

DATA SCIENTIST | AI/ML ENGINEER | MLOPS SPECIALIST
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PROFFESIONAL SUMMARY

Data Scientist, AI/ML Engineer, and MLOps Specialist with 5+ years of experience applying advanced analytics, machine learning, and deep learning techniques to solve real-world business challenges. Skilled in designing, developing, and operationalizing end-to-end AI/ML solutions across healthcare, finance, energy, and retail. I am proficient in Python, SQL, and R for feature engineering, predictive modeling, NLP, deep learning, and time series forecasting, with hands-on expertise in TensorFlow, PyTorch, and Scikit-learn. Experienced in building scalable data architectures and pipelines using Spark, Kafka, and Airflow, and deploying enterprise-grade AI systems on AWS, Azure, and GCP. Adept in implementing MLOps best practices using Docker, Kubernetes, MLflow, and CI/CD automation to ensure scalability, reliability, and governance. Recognized for bridging analytics, AI engineering, and MLOps to deliver intelligent, automated, and business-impactful AI solutions at scale.

CORE COMPETENCIES & TECHNICAL SKILLS

Core Competencies: Data Science, Advanced Analytics, Machine Learning, Deep Learning, NLP, Big Data, Business Intelligence, Cloud Computing, Data Engineering, MLOps

Programming & Scripting: Python, SQL, R, Scala, Java, Bash

Data Analysis & Visualization: Power BI, Tableau, Looker, Qlik Sense, Excel, Matplotlib, Seaborn, Plotly

Statistical & Analytical Methods: Descriptive & Inferential Statistics, Hypothesis Testing, A/B Testing, Regression Analysis, Forecasting (ARIMA, Prophet, GARCH), Feature Engineering

Machine Learning & AI: Scikit-learn, XGBoost, LightGBM, PyCaret, Statsmodels, TensorFlow, Keras, PyTorch, CNN, RNN, LSTM, GANs, Transformers, MLflow

Natural Language Processing (NLP): spaCy, NLTK, BERT, GPT, Word2Vec, Gensim, Sentiment Analysis, Text Classification, Topic Modeling

Big Data & ETL Tools: Apache Spark, Hadoop (HDFS, Hive, MapReduce), Apache Kafka, Databricks, Apache NiFi, Apache Airflow, dbt, Talend, Informatica, Alteryx

Cloud Platforms: AWS (S3, EC2, EMR, SageMaker, Redshift), Azure (Data Lake, Synapse, Azure ML), GCP (BigQuery, Dataflow, Vertex AI)

Databases & Data Warehousing: MySQL, PostgreSQL, Oracle, MongoDB, Redis, Snowflake, Google BigQuery, Azure Synapse Analytics

MLOps & Deployment: Docker, Kubernetes, Jenkins, Git/GitHub/Bitbucket, FastAPI, Flask, CI/CD Pipelines, MLflow, Weights & Biases

PROFESSIONAL EXPERIENCE

Cheniere Energy | Houston, Texas (Remote)
Data Science & MLOps Specialist | Aug 2024 – Present

Cheniere Energy is a global leader in liquefied natural gas (LNG) production and export. My role focuses on designing, deploying, and operationalizing AI/ML solutions for predictive maintenance, production optimization, and real-time monitoring. I apply MLOps best practices to build scalable, automated, and compliant AI pipelines that enhance operational reliability and efficiency across LNG facilities.

- Designed and deployed end-to-end ML pipelines using Python, TensorFlow, and Scikit-learn for predictive maintenance and production optimization, reducing model training time by 30%.
- Built deep learning models using CNNs, Autoencoders, and LSTMs to detect anomalies in IoT sensor data, improving early fault detection accuracy by 26%.
- Developed real-time data streaming pipelines with Apache Kafka and Spark Structured Streaming to support continuous process monitoring.
- Created digital twin simulations using Python and Databricks to analyze compressor unit behavior, improving reliability and scheduling.
- Implemented centralized feature stores and model registries in MLflow to streamline model tracking and reproducibility.
- Automated CI/CD workflows with Docker, Kubernetes, and GitHub Actions, reducing deployment cycles from weeks to hours.
- Deployed scalable ML workloads using AWS SageMaker and Azure ML, achieving 99% uptime in production environments.
- Applied explainable AI (SHAP) for model interpretability, enabling actionable insights for operational teams.
- Collaborated with SCADA engineers to integrate predictive insights into control systems for real-time decision automation.

Impact/Outcomes: Enhanced production reliability and reduced unplanned downtime by integrating predictive models into LNG operations. Improved scalability, governance, and automation of AI systems—accelerating Cheniere’s digital transformation and operational excellence initiatives.

Sevita Health | Boston, Massachusetts
Applied Data Science Analyst | Aug 2023 – Jul 2024

Sevita Health is a nationwide healthcare organization providing specialized, outcome-driven care. My role involved applying data science, predictive analytics, and NLP to improve patient engagement, optimize scheduling, and support clinical decision-making while ensuring full HIPAA compliance.

- Developed predictive analytics models (XGBoost, LightGBM, Logistic Regression) using EHR and behavioral data to predict patient appointment no-shows, reducing missed visits by 18%.
- Engineered complex features—such as chronic conditions, prior cancellations, and appointments lead times—to enhance model precision and interpretability.
- Built NLP pipelines using spaCy and BERT to extract ICD codes and clinical terms for automated compliance audits.
- Created Power BI dashboards for operations teams to visualize attendance trends, outreach performance, and patient segmentation.
- Designed ETL workflows in Spark and SQL to integrate structured and unstructured data, ensuring high-quality inputs for modeling.
- Automated ML retraining and validation pipelines with MLflow and Azure ML to maintain model accuracy.
- Applied explainable AI (SHAP, LIME) to interpret predictions and improve clinician trust in AI-assisted workflows.
- Conducted A/B testing to optimize communication strategies, increasing patient engagement rates by 12%.

- Partnered with clinical and compliance teams to deploy models securely within HIPAA-aligned governance frameworks.

Impact/Outcomes: Reduced appointment no-shows, improved resource utilization, and increased patient engagement through data-driven outreach. Delivered scalable, interpretable, and compliant AI solutions that enhanced Sevita's operational and clinical efficiency.

Plural Technologies | Bengaluru, India

Data Science & Analytics Specialist | Dec 2019 – Aug 2022

Plural Technologies is a data consulting firm delivering AI/ML, analytics, and business intelligence solutions across finance, retail, and insurance sectors. My role focused on building predictive models, developing scalable data pipelines, and partnering with client teams to deploy actionable insights that improved performance and reduced business risk.

- Developed fraud detection and risk-scoring models using ensemble and deep learning techniques, reducing financial transaction fraud exposure by 22%.
- Built churn prediction pipelines using XGBoost and Random Forest for retail clients, improving retention campaign ROI.
- Created insurance claim severity and loss prediction models leveraging regression and survival analysis for pricing optimization.
- Delivered retail demand forecasting models using ARIMA, Prophet, and LSTM to reduce overstock and optimize inventory planning.
- Designed sentiment analysis systems using NLP tools (spaCy, BERT, Gensim) to extract actionable insights from customer feedback.
- Implemented real-time credit monitoring and fraud flagging solutions with Spark Streaming and Kafka.
- Automated ETL workflows and data integration across APIs, relational systems, and cloud warehouses (AWS, Azure, GCP).
- Performed portfolio risk and stress testing for banking clients using statistical modeling and capital adequacy analysis.
- Created BI dashboards in Power BI and Tableau to visualize KPIs, financial metrics, and operational insights.
- Delivered analytics workshops and training sessions to upskill client teams in adopting AI and data-driven strategies.

Impact/Outcomes: Enabled clients across finance, retail, and insurance industries to mitigate financial risk, improve customer engagement, and operationalize data-driven decision frameworks. Delivered measurable ROI through predictive analytics, automation, and scalable AI solutions.

EDUCATION: Master's in computer science – *Wright State University*